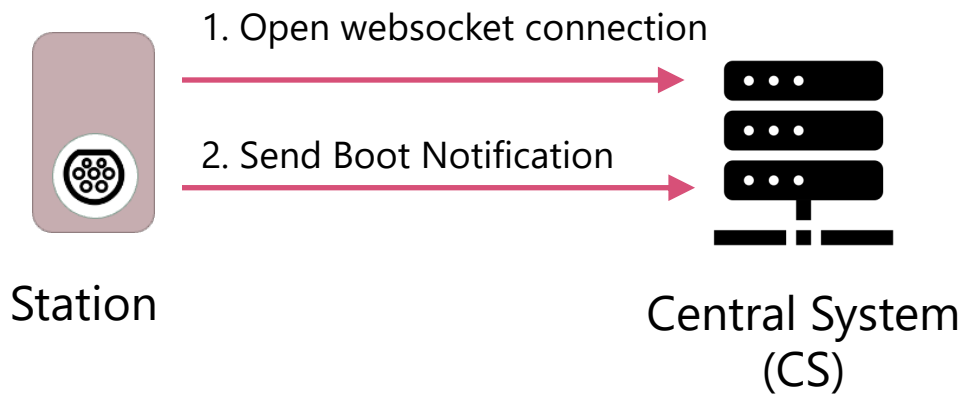


10. RECAP - WHAT DID WE LEARN?



Connecting Stations to a Central System

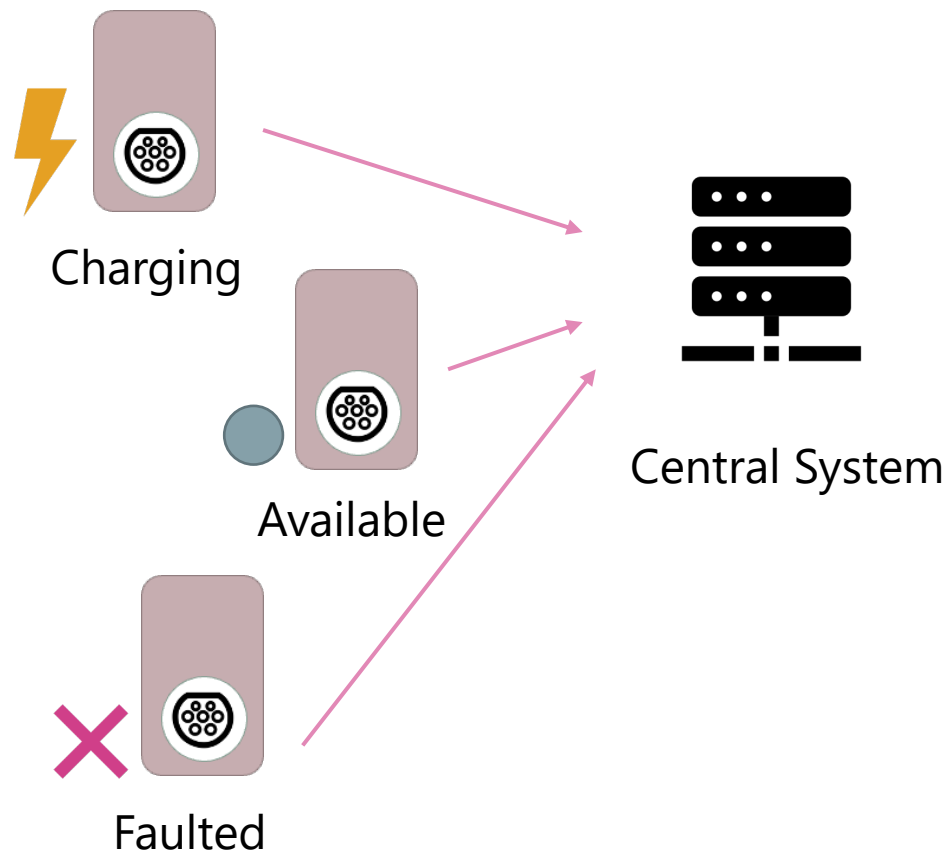
When a station **is powered on**, it should automatically **connect to the central system** in the following way::



1. The station checks its internal configuration for the URL of the central system and tries to open a new **WebSocket** connection there.
2. After the connection is successfully established, the station sends a '**Boot Notification**' message to the central system, providing details about the station that is now opening the connection.



Charging Station Status



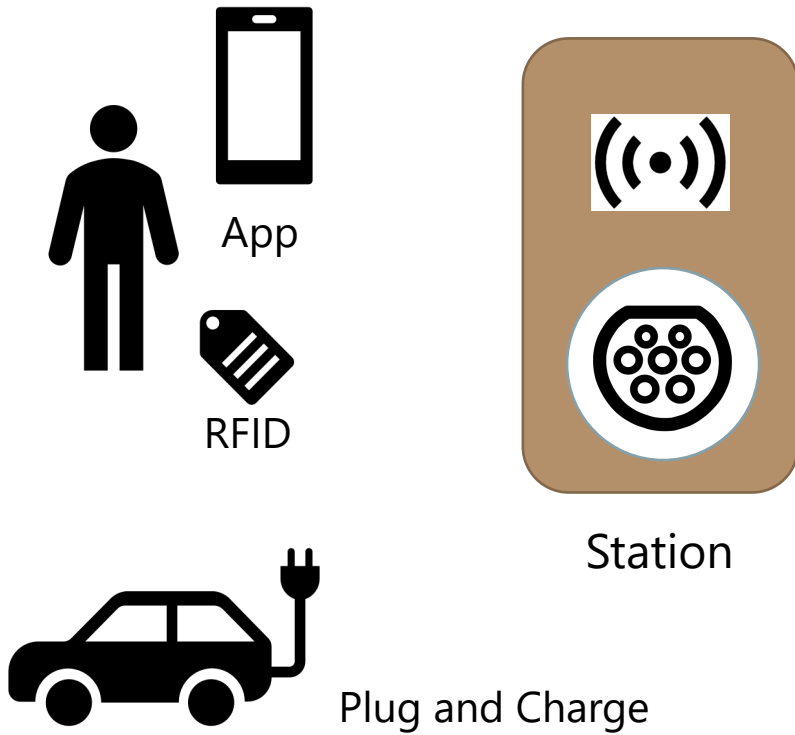
Charging stations can have different **operational statuses**. A status is "**Available**" when a station is free and nobody is using it, "**Charging**" when there is an EV charging at the station, or "**Faulted**" if there is some error on the station.

A status is always **linked to a single connector** of a station. For example, if a station has two connectors, one may be "Available," while another is "Charging."

Stations report statuses to the central system with a message called "**Status Notification.**"



Different ways to use stations



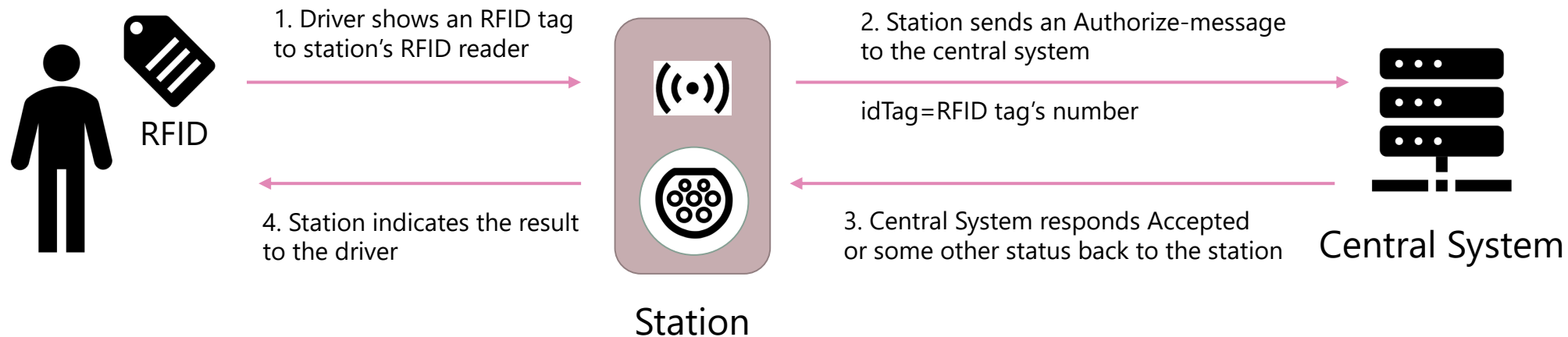
When a driver wants to use a charging station, the first step is often to **authorize** the driver. This is done to identify the driver and ensure that they are allowed to use the station.

There are **several ways** to authorize drivers:

1. **RFID tokens** – Drivers can show an RFID token to the station's RFID reader to authorize themselves.
2. **Mobile apps** – Drivers can use mobile apps to authenticate themselves and use a station.
3. **Plug and Charge** – Drivers can also be authorized through the charging cable, provided that both the EV and the station support the ISO 15118 standard.



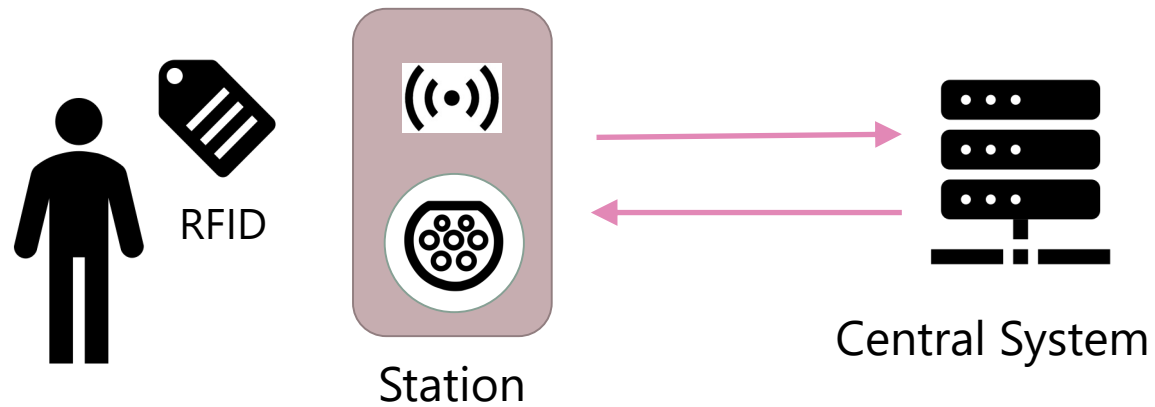
Basic Authorization Flow



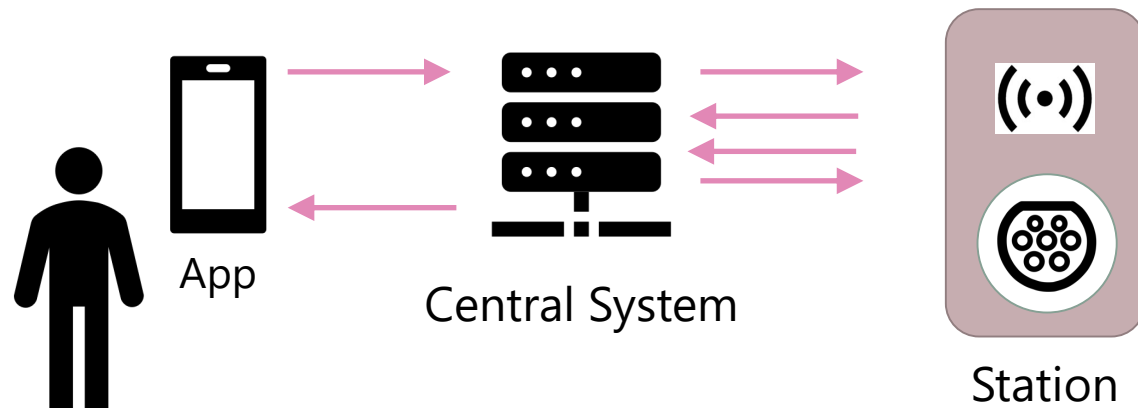
The simplest authorization flow is through the use of **an RFID tag**. In this case, a driver shows the RFID tag to the station, which then sends an **Authorize** message with the RFID tag's number to the Central System. The Central System **maintains a list of accepted RFID tags** and checks to see if the provided RFID tag is in the list.



Mobile Apps and other UIs



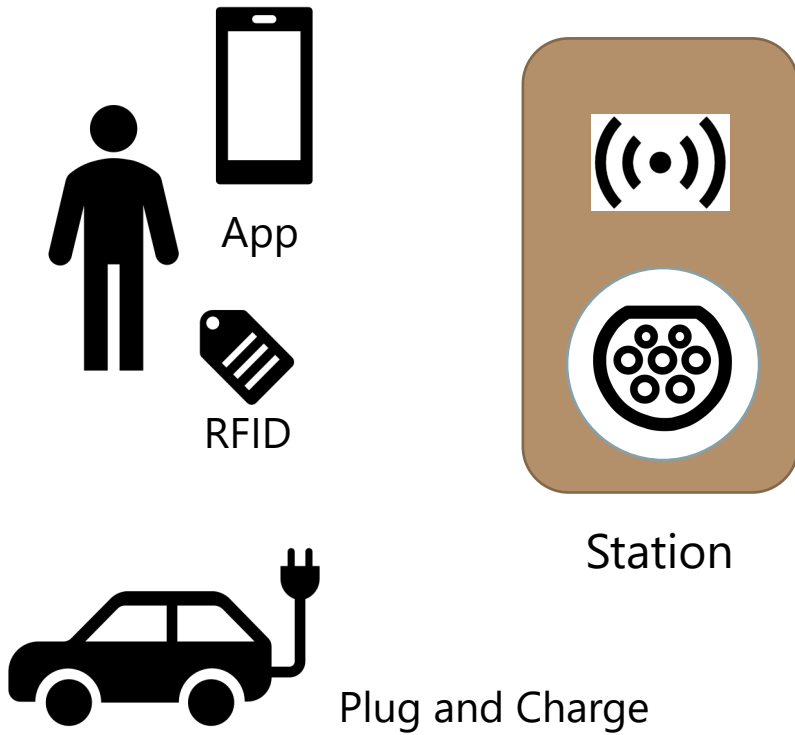
In the previous lesson, we discussed the basic charging flow when an RFID tag is used to start charging at a station: the station sends **Authorize** and **StartTransaction** messages to the central system, which then accepts or rejects them.



Another way to use stations is to start a charge **through a mobile app** or a remote interface such as **an admin system**. In these cases, the flow is slightly different: the central system initiates the charging with a **RemoteStartTransaction**.



Different ways to stop charging

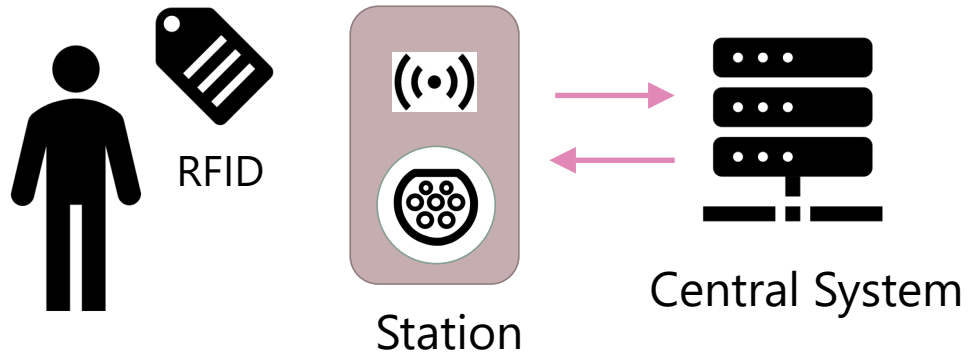


There are several ways to **stop** a charging transaction, including:

1. **RFID tokens** – Showing the same RFID tag that was used to start the charge will usually stop the transaction.
2. **Mobile apps** – Using the stop button in the app will end the charge.
3. **Plug and Charge** - Disconnecting the cable from the EV will stop the transaction.

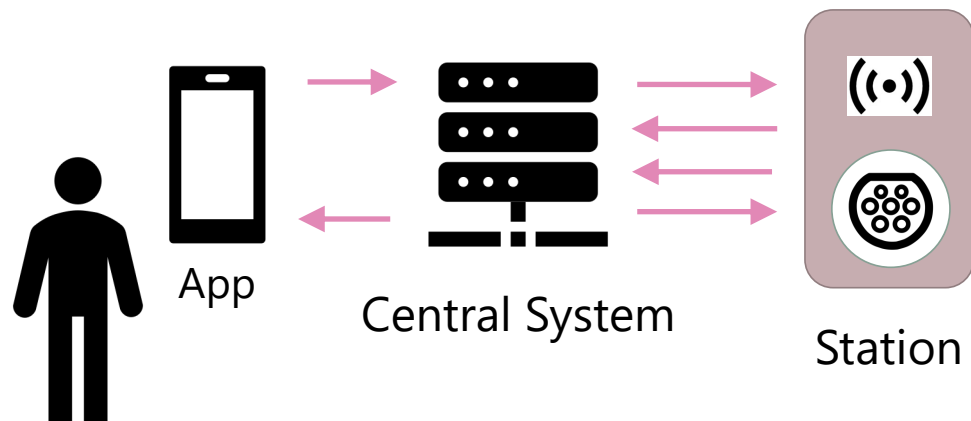


Summary: Remote vs. Local stop



Local stop flow:

1. An EV driver presents their RFID tag to the charging station.
2. The station sends a "**StopTransaction.req**" message to the central system.
3. The central system responds with a "**StopTransaction.conf**" message, indicating that it has received the stop message.

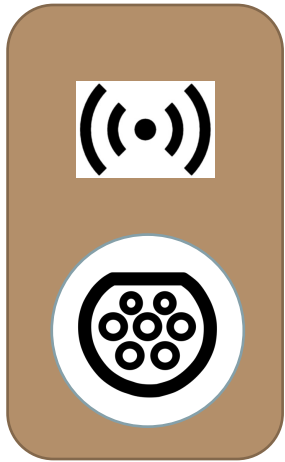


Remote stop flow:

1. An EV driver stops charging from an app. The app calls an API in the central system, which checks that stopping is allowed.
2. The central system sends a "**RemoteStopTransaction.req**" message to the charging station, which either accepts or rejects the request.
3. If the station accepts the remote stop, it sends a "StopTransaction.req" message and central system confirms it



Further studies



Station

If you want to continue learning other OCPP messages and use cases, we recommend going through the protocol documentation.

Especially the following messages may be useful to learn next:

1. Reset
2. GetConfiguration and ChangeConfiguration
3. Change Availability
4. Update Firmware
5. Unlock Connector

